

Solução do Problema PILOT - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: PILOT
- Número de Variáveis: 3652
- Número de Restrições: 1441
- Inviabilidade Primal: 3.227e-08
- Inviabilidade Dual: 0.000e+00
- Valor Primal: -5.575e+02
- Valor Dual: 6.741e+01
- Gap: -5.135e-17
- Número de Iterações: 40

2 Variáveis Primais e Custos Reduzidos

Tabela 1: Variáveis primais (x) e custos reduzidos (z) do problema PILOT

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	8.500e+02	1	1.804e-02
2	4.690e+02	2	5.655e-02
3	1.288e+03	3	2.386e-02
4	1.019e+03	4	0.000e+00
5	7.190e+02	5	5.989e-02
6	8.690e+02	6	7.705e-04
7	5.340e+02	7	8.535e-04
8	1.483e+03	8	0.000e+00
9	1.191e+03	9	0.000e+00
10	8.429e+02	10	0.000e+00
11	1.042e+03	11	0.000e+00
12	5.340e+02	12	1.509e-04

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
13	1.553e+03	13	0.000e+00
14	1.378e+03	14	0.000e+00
15	1.034e+03	15	0.000e+00
16	1.325e+03	16	0.000e+00
17	5.340e+02	17	3.877e-04
18	1.826e+03	18	0.000e+00
19	1.553e+03	19	0.000e+00
20	8.240e+02	20	3.967e-07
21	8.690e+02	21	1.247e-06
22	9.290e+02	22	0.000e+00
23	2.024e+03	23	0.000e+00
24	2.030e+03	24	0.000e+00
25	8.240e+02	25	5.059e-06
26	8.690e+02	26	3.518e-06
27	5.340e+02	27	8.612e-06
28	2.262e+03	28	0.000e+00
29	2.218e+03	29	0.000e+00
30	1.824e+03	30	0.000e+00
31	1.507e+03	31	0.000e+00
32	7.225e+02	32	0.000e+00
33	2.458e+03	33	0.000e+00
34	2.367e+03	34	0.000e+00
35	1.541e+03	35	0.000e+00
36	8.690e+02	36	8.754e-06
37	5.340e+02	37	1.352e-05
38	2.657e+03	38	0.000e+00
39	2.134e+03	39	0.000e+00
40	4.085e+03	40	0.000e+00
41	8.690e+02	41	1.518e-05
42	1.716e+03	42	0.000e+00
43	3.653e+03	43	0.000e+00
44	2.750e+03	44	0.000e+00
45	2.991e+03	45	0.000e+00
46	1.500e+02	46	3.111e-01
47	2.560e+02	47	1.009e-01
48	4.520e+02	48	2.417e-01
49	1.570e+02	49	1.670e-02
50	2.970e+02	50	1.184e-02
51	5.250e+02	51	1.639e-02
52	1.690e+02	52	1.311e-02
53	3.280e+02	53	1.177e-02
54	5.950e+02	54	1.285e-02
55	1.820e+02	55	1.053e-02
56	3.630e+02	56	9.683e-03
57	6.580e+02	57	1.033e-02
58	1.970e+02	58	8.233e-03
59	4.010e+02	59	7.652e-03
60	7.270e+02	60	8.024e-03
61	2.120e+02	61	6.370e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
62	4.430e+02	62	5.950e-03
63	8.030e+02	63	6.186e-03
64	2.280e+02	64	3.703e-03
65	4.900e+02	65	3.308e-03
66	8.880e+02	66	3.711e-03
67	2.460e+02	67	3.993e-03
68	5.410e+02	68	3.784e-03
69	9.810e+02	69	3.867e-03
70	2.650e+02	70	8.366e-03
71	5.980e+02	71	7.528e-03
72	1.084e+03	72	8.139e-03
73	1.339e+02	73	0.000e+00
74	5.802e-02	74	0.000e+00
75	2.135e+01	75	0.000e+00
76	7.789e+01	76	0.000e+00
77	3.147e+01	77	0.000e+00
78	3.393e+02	78	0.000e+00
79	1.095e+02	79	0.000e+00
80	1.094e+01	80	0.000e+00
81	8.886e+01	81	0.000e+00
82	2.354e+02	82	0.000e+00
83	3.867e+01	83	0.000e+00
84	1.297e+02	84	0.000e+00
85	2.744e+01	85	0.000e+00
86	1.297e+02	86	0.000e+00
87	5.974e+01	87	0.000e+00
88	3.360e+01	88	0.000e+00
89	9.775e+01	89	0.000e+00
90	1.451e+02	90	0.000e+00
91	1.893e+02	91	0.000e+00
92	1.202e+02	92	0.000e+00
93	4.055e+02	93	0.000e+00
94	5.499e+01	94	0.000e+00
95	9.945e+01	95	0.000e+00
96	3.891e+02	96	0.000e+00
97	1.095e+02	97	0.000e+00
98	1.094e+01	98	0.000e+00
99	8.886e+01	99	0.000e+00
100	2.354e+02	100	0.000e+00
101	3.867e+01	101	0.000e+00
102	1.297e+02	102	0.000e+00
103	2.744e+01	103	0.000e+00
104	1.297e+02	104	0.000e+00
105	5.974e+01	105	0.000e+00
106	3.360e+01	106	0.000e+00
107	9.775e+01	107	0.000e+00
108	1.451e+02	108	0.000e+00
109	1.893e+02	109	0.000e+00
110	1.202e+02	110	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
111	4.055e+02	111	0.000e+00
112	5.499e+01	112	0.000e+00
113	9.945e+01	113	0.000e+00
114	3.891e+02	114	0.000e+00
115	2.064e+02	115	0.000e+00
116	2.064e+02	116	0.000e+00
117	5.602e+01	117	0.000e+00
118	1.428e+01	118	1.105e-01
119	9.680e-01	119	1.337e-01
120	8.083e+00	120	1.556e-01
121	2.488e+00	121	1.390e-01
122	1.051e+01	122	2.107e-01
123	8.910e-01	123	1.255e-01
124	3.408e+00	124	1.364e-01
125	1.736e+00	125	1.166e-01
126	1.293e+00	126	1.567e-01
127	1.936e+01	127	1.128e-01
128	1.521e+01	128	1.127e-01
129	1.645e+01	129	1.386e-01
130	1.131e+01	130	1.361e-01
131	1.640e+01	131	1.650e-01
132	0.000e+00	132	1.105e-01
133	0.000e+00	133	1.337e-01
134	0.000e+00	134	1.556e-01
135	0.000e+00	135	1.390e-01
136	0.000e+00	136	2.107e-01
137	0.000e+00	137	1.255e-01
138	0.000e+00	138	1.364e-01
139	0.000e+00	139	1.166e-01
140	0.000e+00	140	1.567e-01
141	0.000e+00	141	1.128e-01
142	0.000e+00	142	1.127e-01
143	0.000e+00	143	1.386e-01
144	0.000e+00	144	1.361e-01
145	0.000e+00	145	1.650e-01
146	3.516e+00	146	-1.105e-01
147	2.294e+00	147	-1.337e-01
148	1.011e+01	148	-1.556e-01
149	3.430e+00	149	-1.390e-01
150	5.877e+00	150	-2.107e-01
151	1.376e+00	151	-1.255e-01
152	1.113e+01	152	-1.364e-01
153	5.103e+00	153	-1.166e-01
154	3.577e+00	154	-1.567e-01
155	6.459e+00	155	-1.128e-01
156	1.822e+01	156	-1.127e-01
157	1.672e+01	157	-1.386e-01
158	2.613e+00	158	-1.361e-01
159	1.271e+02	159	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
160	1.356e+02	160	0.000e+00
161	1.709e+02	161	0.000e+00
162	7.371e-01	162	0.000e+00
163	1.820e+01	163	0.000e+00
164	7.177e+01	164	0.000e+00
165	5.007e+01	165	0.000e+00
166	3.903e+02	166	0.000e+00
167	1.252e+02	167	0.000e+00
168	1.295e+01	168	0.000e+00
169	1.006e+02	169	0.000e+00
170	2.613e+02	170	0.000e+00
171	4.479e+01	171	0.000e+00
172	1.541e+02	172	0.000e+00
173	3.096e+01	173	0.000e+00
174	1.498e+02	174	0.000e+00
175	6.886e+01	175	0.000e+00
176	3.710e+01	176	0.000e+00
177	1.282e+02	177	0.000e+00
178	1.723e+02	178	0.000e+00
179	2.216e+02	179	0.000e+00
180	1.389e+02	180	0.000e+00
181	4.745e+02	181	0.000e+00
182	6.986e+01	182	0.000e+00
183	1.202e+02	183	0.000e+00
184	4.594e+02	184	0.000e+00
185	1.030e+02	185	-2.730e-02
186	1.019e+01	186	-1.365e-02
187	7.552e+01	187	-1.145e-02
188	2.100e+02	188	-1.015e-02
189	3.909e+01	189	0.000e+00
190	1.183e+02	190	-2.377e-02
191	2.457e+01	191	-3.656e-02
192	1.091e+02	192	-1.946e-02
193	5.269e+01	193	-1.580e-02
194	2.947e+01	194	-1.820e-02
195	8.380e+01	195	0.000e+00
196	1.185e+02	196	-2.035e-02
197	1.650e+02	197	-2.061e-02
198	9.559e+01	198	-2.046e-02
199	3.459e+02	199	-6.198e-03
200	4.787e+01	200	-5.963e-02
201	9.022e+01	201	-6.906e-03
202	3.252e+02	202	-5.738e-03
203	2.222e+01	203	0.000e+00
204	2.759e+00	204	0.000e+00
205	2.511e+01	205	0.000e+00
206	5.130e+01	206	0.000e+00
207	5.700e+00	207	4.298e-02
208	3.582e+01	208	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
209	6.394e+00	209	0.000e+00
210	4.066e+01	210	0.000e+00
211	1.617e+01	211	0.000e+00
212	7.630e+00	212	0.000e+00
213	4.440e+01	213	1.397e-02
214	5.381e+01	214	0.000e+00
215	5.657e+01	215	0.000e+00
216	4.332e+01	216	0.000e+00
217	1.286e+02	217	0.000e+00
218	2.199e+01	218	0.000e+00
219	2.996e+01	219	0.000e+00
220	0.000e+00	220	8.309e-03
221	1.341e+02	221	0.000e+00
222	2.505e+02	222	0.000e+00
223	2.045e+02	223	-3.181e-02
224	4.600e+01	224	9.594e-02
225	6.605e+01	225	0.000e+00
226	1.873e+01	226	2.452e-02
227	2.064e+00	227	1.260e-02
228	1.326e+01	228	2.175e-02
229	3.264e+00	229	1.418e-02
230	1.637e+01	230	3.774e-02
231	1.169e+00	231	5.252e-02
232	5.030e+00	232	1.980e-02
233	4.413e+00	233	4.047e-02
234	3.287e+00	234	7.367e-02
235	3.201e+01	235	0.000e+00
236	1.995e+01	236	2.556e-02
237	2.159e+01	237	2.693e-02
238	1.299e+01	238	4.262e-02
239	1.882e+01	239	1.510e-02
240	0.000e+00	240	5.280e-02
241	0.000e+00	241	2.809e-02
242	0.000e+00	242	4.882e-02
243	0.000e+00	243	3.641e-02
244	0.000e+00	244	6.259e-02
245	0.000e+00	245	8.139e-02
246	0.000e+00	246	4.542e-02
247	0.000e+00	247	4.805e-02
248	0.000e+00	248	8.912e-02
249	0.000e+00	249	1.714e-02
250	0.000e+00	250	4.602e-02
251	0.000e+00	251	4.456e-02
252	0.000e+00	252	6.847e-02
253	0.000e+00	253	4.095e-02
254	4.219e+00	254	-2.452e-02
255	3.212e+00	255	-1.260e-02
256	1.415e+01	256	-2.175e-02
257	4.116e+00	257	-1.418e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
258	8.816e+00	258	-3.774e-02
259	2.064e+00	259	-5.252e-02
260	1.558e+01	260	-1.980e-02
261	1.021e+01	261	-4.047e-02
262	7.512e+00	262	-7.367e-02
263	1.112e+01	263	0.000e+00
264	2.550e+01	264	-2.556e-02
265	2.843e+01	265	-2.693e-02
266	3.658e+00	266	-4.262e-02
267	1.672e+02	267	0.000e+00
268	1.948e+02	268	0.000e+00
269	1.885e+02	269	0.000e+00
270	1.420e+01	270	0.000e+00
271	3.151e+01	271	0.000e+00
272	1.159e+02	272	0.000e+00
273	6.032e+01	273	0.000e+00
274	4.858e+02	274	0.000e+00
275	1.775e+02	275	0.000e+00
276	1.945e+01	276	0.000e+00
277	1.348e+02	277	0.000e+00
278	3.563e+02	278	0.000e+00
279	5.688e+01	279	0.000e+00
280	2.037e+02	280	0.000e+00
281	4.285e+01	281	0.000e+00
282	2.010e+02	282	0.000e+00
283	5.369e+01	283	0.000e+00
284	2.740e+01	284	0.000e+00
285	1.025e+02	285	0.000e+00
286	1.655e+02	286	0.000e+00
287	2.156e+02	287	0.000e+00
288	1.772e+02	288	0.000e+00
289	5.784e+02	289	0.000e+00
290	9.056e+01	290	0.000e+00
291	1.411e+02	291	0.000e+00
292	5.508e+02	292	0.000e+00
293	6.911e+01	293	-8.445e-03
294	6.837e+00	294	-1.677e-03
295	5.067e+01	295	-1.017e-03
296	1.409e+02	296	-3.480e-03
297	2.751e+01	297	-1.315e-02
298	7.934e+01	298	-6.815e-03
299	1.648e+01	299	-1.125e-02
300	7.319e+01	300	-7.153e-03
301	3.535e+01	301	-4.344e-03
302	1.977e+01	302	-3.249e-03
303	5.809e+01	303	-1.127e-02
304	7.950e+01	304	-1.361e-02
305	1.107e+02	305	-1.794e-02
306	6.413e+01	306	-7.194e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
307	2.321e+02	307	-4.768e-03
308	3.212e+01	308	-1.971e-02
309	6.053e+01	309	-4.412e-03
310	2.182e+02	310	-5.099e-03
311	8.617e+01	311	0.000e+00
312	9.852e+00	312	0.000e+00
313	5.901e+01	313	0.000e+00
314	1.642e+02	314	0.000e+00
315	0.000e+00	315	7.827e-04
316	2.367e+01	316	0.000e+00
317	8.851e+01	317	0.000e+00
318	1.997e+01	318	0.000e+00
319	8.719e+01	319	0.000e+00
320	2.175e+00	320	0.000e+00
321	0.000e+00	321	2.308e-03
322	0.000e+00	322	5.726e-04
323	0.000e+00	323	1.316e-03
324	3.218e+01	324	0.000e+00
325	4.831e+01	325	0.000e+00
326	6.974e+01	326	0.000e+00
327	2.177e+02	327	0.000e+00
328	0.000e+00	328	2.495e-03
329	3.646e+01	329	0.000e+00
330	5.060e+01	330	0.000e+00
331	0.000e+00	331	1.373e-03
332	1.984e+02	332	0.000e+00
333	0.000e+00	333	1.513e-03
334	2.750e+02	334	0.000e+00
335	1.945e+02	335	-2.739e-02
336	3.667e+01	336	0.000e+00
337	8.029e+01	337	0.000e+00
338	2.654e+01	338	-1.527e-02
339	2.925e+00	339	-2.133e-02
340	1.878e+01	340	-1.452e-02
341	0.000e+00	341	0.000e+00
342	2.320e+01	342	-1.289e-02
343	1.657e+00	343	-5.570e-03
344	7.128e+00	344	-1.230e-02
345	0.000e+00	345	1.814e-03
346	0.000e+00	346	0.000e+00
347	0.000e+00	347	0.000e+00
348	0.000e+00	348	8.149e-03
349	0.000e+00	349	7.754e-03
350	1.841e+01	350	-1.515e-02
351	2.667e+01	351	-1.291e-02
352	0.000e+00	352	1.847e-02
353	1.358e+00	353	-2.857e-03
354	0.000e+00	354	1.777e-02
355	0.000e+00	355	2.651e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
356	0.000e+00	356	1.675e-02
357	0.000e+00	357	2.887e-02
358	0.000e+00	358	1.826e-02
359	0.000e+00	359	9.698e-03
360	0.000e+00	360	1.843e-02
361	0.000e+00	361	1.818e-02
362	0.000e+00	362	2.713e-02
363	0.000e+00	363	2.644e-02
364	0.000e+00	364	1.568e-02
365	0.000e+00	365	1.793e-02
366	0.000e+00	366	1.527e-02
367	0.000e+00	367	2.133e-02
368	0.000e+00	368	1.452e-02
369	1.534e+00	369	0.000e+00
370	0.000e+00	370	1.289e-02
371	0.000e+00	371	5.570e-03
372	0.000e+00	372	1.230e-02
373	1.531e+01	373	-1.814e-03
374	9.311e+00	374	0.000e+00
375	2.376e+01	375	0.000e+00
376	4.554e+01	376	-8.149e-03
377	4.849e+01	377	-7.754e-03
378	0.000e+00	378	1.515e-02
379	1.372e+02	379	0.000e+00
380	1.649e+02	380	0.000e+00
381	2.306e+02	381	0.000e+00
382	1.753e+01	382	0.000e+00
383	4.028e+01	383	0.000e+00
384	1.290e+02	384	0.000e+00
385	6.794e+01	385	0.000e+00
386	5.871e+02	386	0.000e+00
387	2.162e+02	387	0.000e+00
388	2.174e+01	388	0.000e+00
389	1.579e+02	389	0.000e+00
390	4.439e+02	390	0.000e+00
391	4.802e+01	391	0.000e+00
392	1.897e+02	392	0.000e+00
393	4.507e+01	393	0.000e+00
394	1.728e+02	394	0.000e+00
395	8.889e+01	395	0.000e+00
396	3.572e+01	396	0.000e+00
397	2.719e+02	397	0.000e+00
398	3.165e+02	398	0.000e+00
399	2.476e+02	399	0.000e+00
400	2.118e+02	400	0.000e+00
401	6.580e+02	401	0.000e+00
402	1.067e+02	402	0.000e+00
403	1.645e+02	403	0.000e+00
404	6.299e+02	404	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
405	3.782e+01	405	-7.929e-03
406	3.741e+00	406	-2.501e-03
407	2.772e+01	407	-1.518e-03
408	7.708e+01	408	-3.210e-03
409	1.505e+01	409	-1.180e-02
410	4.341e+01	410	-7.054e-03
411	9.019e+00	411	-1.135e-02
412	4.005e+01	412	-6.386e-03
413	1.934e+01	413	-4.312e-03
414	1.082e+01	414	-4.451e-03
415	0.000e+00	415	1.198e-03
416	4.350e+01	416	-2.412e-03
417	6.058e+01	417	-4.158e-04
418	3.509e+01	418	-6.581e-03
419	1.270e+02	419	-3.070e-03
420	1.757e+01	420	-1.911e-02
421	3.312e+01	421	-2.744e-03
422	1.194e+02	422	-2.472e-03
423	7.331e+01	423	0.000e+00
424	5.806e+00	424	0.000e+00
425	4.980e+01	425	0.000e+00
426	1.591e+02	426	0.000e+00
427	0.000e+00	427	7.379e-04
428	4.447e+00	428	0.000e+00
429	2.731e+01	429	0.000e+00
430	1.064e+01	430	0.000e+00
431	1.101e+01	431	0.000e+00
432	5.363e+01	432	0.000e+00
433	0.000e+00	433	1.880e-03
434	1.842e+01	434	0.000e+00
435	0.000e+00	435	1.186e-02
436	2.341e+02	436	0.000e+00
437	0.000e+00	437	9.346e-03
438	1.951e+02	438	0.000e+00
439	0.000e+00	439	1.507e-02
440	9.062e+01	440	0.000e+00
441	7.012e+01	441	0.000e+00
442	0.000e+00	442	3.053e-03
443	2.039e+02	443	0.000e+00
444	3.396e+01	444	0.000e+00
445	0.000e+00	445	3.104e-03
446	5.528e+01	446	0.000e+00
447	0.000e+00	447	4.706e-03
448	1.981e+02	448	0.000e+00
449	3.383e+02	449	0.000e+00
450	1.850e+02	450	-1.938e-02
451	7.681e+01	451	0.000e+00
452	9.250e+01	452	0.000e+00
453	3.084e+01	453	-1.536e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
454	3.398e+00	454	-7.495e-03
455	2.183e+01	455	-1.726e-03
456	0.000e+00	456	1.000e-02
457	0.000e+00	457	0.000e+00
458	0.000e+00	458	7.702e-03
459	0.000e+00	459	4.211e-04
460	0.000e+00	460	7.818e-03
461	0.000e+00	461	1.071e-02
462	4.445e+01	462	-2.610e-03
463	3.286e+01	463	-8.784e-04
464	0.000e+00	464	1.262e-03
465	2.139e+01	465	-2.053e-03
466	3.100e+01	466	-2.902e-04
467	0.000e+00	467	1.904e-02
468	0.000e+00	468	3.774e-03
469	0.000e+00	469	1.797e-02
470	0.000e+00	470	2.617e-02
471	0.000e+00	471	1.808e-02
472	0.000e+00	472	2.871e-02
473	0.000e+00	473	1.906e-02
474	0.000e+00	474	1.284e-02
475	0.000e+00	475	2.196e-02
476	0.000e+00	476	7.092e-03
477	0.000e+00	477	1.070e-02
478	0.000e+00	478	1.124e-02
479	0.000e+00	479	1.675e-02
480	0.000e+00	480	1.852e-02
481	0.000e+00	481	1.536e-03
482	0.000e+00	482	7.495e-03
483	0.000e+00	483	1.726e-03
484	9.947e+00	484	-1.000e-02
485	1.163e+01	485	0.000e+00
486	3.990e+00	486	-7.702e-03
487	3.228e+01	487	-4.211e-04
488	2.143e+01	488	-7.818e-03
489	1.502e+01	489	-1.071e-02
490	0.000e+00	490	2.610e-03
491	0.000e+00	491	8.784e-04
492	8.193e+01	492	-1.262e-03
493	0.000e+00	493	2.053e-03
494	1.762e+02	494	0.000e+00
495	2.038e+02	495	0.000e+00
496	2.761e+02	496	0.000e+00
497	1.778e+01	497	0.000e+00
498	4.697e+01	498	0.000e+00
499	1.405e+02	499	0.000e+00
500	7.984e+01	500	0.000e+00
501	6.681e+02	501	0.000e+00
502	2.013e+02	502	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
503	2.657e+01	503	0.000e+00
504	1.812e+02	504	0.000e+00
505	3.767e+02	505	0.000e+00
506	5.792e+01	506	0.000e+00
507	2.885e+02	507	0.000e+00
508	5.105e+01	508	0.000e+00
509	2.946e+02	509	0.000e+00
510	1.101e+02	510	0.000e+00
511	4.023e+01	511	0.000e+00
512	3.858e+02	512	0.000e+00
513	3.996e+02	513	0.000e+00
514	2.983e+02	514	0.000e+00
515	2.516e+02	515	0.000e+00
516	7.360e+02	516	0.000e+00
517	1.411e+02	517	0.000e+00
518	1.905e+02	518	0.000e+00
519	7.556e+02	519	0.000e+00
520	1.695e+01	520	-3.167e-03
521	1.677e+00	521	-7.662e-04
522	1.243e+01	522	-5.024e-04
523	3.455e+01	523	-1.333e-03
524	6.747e+00	524	-4.610e-03
525	1.946e+01	525	-2.778e-03
526	4.043e+00	526	-4.462e-03
527	1.795e+01	527	-2.624e-03
528	8.671e+00	528	-1.505e-03
529	4.849e+00	529	-5.668e-04
530	0.000e+00	530	1.895e-03
531	1.950e+01	531	-4.557e-04
532	0.000e+00	532	1.111e-03
533	1.573e+01	533	-2.707e-03
534	0.000e+00	534	6.664e-05
535	7.878e+00	535	-7.598e-03
536	0.000e+00	536	6.361e-05
537	0.000e+00	537	8.975e-04
538	2.555e+01	538	0.000e+00
539	9.194e+00	539	0.000e+00
540	5.500e+01	540	0.000e+00
541	1.539e+01	541	0.000e+00
542	0.000e+00	542	7.034e-04
543	2.347e+01	543	0.000e+00
544	1.468e+02	544	0.000e+00
545	1.587e+01	545	0.000e+00
546	1.691e+02	546	0.000e+00
547	3.705e+01	547	0.000e+00
548	0.000e+00	548	1.076e-03
549	1.276e+01	549	0.000e+00
550	0.000e+00	550	9.164e-03
551	1.273e+02	551	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
552	0.000e+00	552	6.689e-03
553	1.281e+02	553	0.000e+00
554	0.000e+00	554	1.154e-02
555	1.355e+02	555	0.000e+00
556	8.266e+01	556	0.000e+00
557	0.000e+00	557	5.317e-03
558	2.763e+02	558	0.000e+00
559	5.618e+01	559	0.000e+00
560	0.000e+00	560	5.075e-03
561	7.569e+01	561	0.000e+00
562	0.000e+00	562	7.520e-03
563	3.151e+02	563	0.000e+00
564	3.898e+02	564	0.000e+00
565	1.760e+02	565	-1.223e-02
566	6.795e+01	566	0.000e+00
567	1.108e+02	567	0.000e+00
568	3.584e+01	568	-3.699e-03
569	3.949e+00	569	-3.839e-03
570	2.536e+01	570	-2.933e-03
571	0.000e+00	571	2.375e-03
572	3.132e+01	572	-2.101e-03
573	0.000e+00	573	2.948e-03
574	9.625e+00	574	-1.635e-03
575	0.000e+00	575	4.379e-03
576	0.000e+00	576	5.712e-03
577	5.165e+01	577	-2.126e-03
578	3.818e+01	578	-2.064e-03
579	0.000e+00	579	0.000e+00
580	2.485e+01	580	-2.393e-03
581	0.000e+00	581	8.813e-04
582	0.000e+00	582	1.092e-02
583	0.000e+00	583	4.163e-03
584	0.000e+00	584	1.105e-02
585	0.000e+00	585	1.386e-02
586	0.000e+00	586	1.074e-02
587	0.000e+00	587	1.786e-02
588	0.000e+00	588	1.160e-02
589	0.000e+00	589	8.145e-03
590	0.000e+00	590	1.370e-02
591	0.000e+00	591	4.764e-03
592	0.000e+00	592	6.159e-03
593	0.000e+00	593	7.084e-03
594	0.000e+00	594	1.096e-02
595	0.000e+00	595	1.424e-02
596	0.000e+00	596	3.699e-03
597	0.000e+00	597	3.839e-03
598	0.000e+00	598	2.933e-03
599	1.269e+01	599	-2.375e-03
600	0.000e+00	600	2.101e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
601	5.091e+00	601	-2.948e-03
602	0.000e+00	602	1.635e-03
603	3.062e+01	603	-4.379e-03
604	2.146e+01	604	-5.712e-03
605	0.000e+00	605	2.126e-03
606	0.000e+00	606	2.064e-03
607	1.003e+02	607	0.000e+00
608	0.000e+00	608	2.393e-03
609	2.037e+02	609	0.000e+00
610	2.314e+02	610	0.000e+00
611	2.726e+02	611	0.000e+00
612	1.914e+01	612	0.000e+00
613	5.169e+01	613	0.000e+00
614	1.410e+02	614	0.000e+00
615	8.058e+01	615	0.000e+00
616	8.047e+02	616	0.000e+00
617	2.201e+02	617	0.000e+00
618	3.019e+01	618	0.000e+00
619	1.977e+02	619	0.000e+00
620	4.040e+02	620	0.000e+00
621	7.988e+01	621	0.000e+00
622	3.205e+02	622	0.000e+00
623	5.480e+01	623	0.000e+00
624	2.927e+02	624	0.000e+00
625	1.293e+02	625	0.000e+00
626	4.349e+01	626	0.000e+00
627	4.641e+02	627	0.000e+00
628	4.489e+02	628	0.000e+00
629	4.221e+02	629	0.000e+00
630	2.992e+02	630	0.000e+00
631	8.548e+02	631	0.000e+00
632	2.304e+02	632	0.000e+00
633	2.193e+02	633	0.000e+00
634	8.743e+02	634	0.000e+00
635	6.520e+00	635	-1.171e-03
636	6.450e-01	636	-1.040e-04
637	4.780e+00	637	-4.921e-05
638	1.329e+01	638	-5.532e-04
639	2.595e+00	639	-1.592e-03
640	7.485e+00	640	-8.666e-04
641	1.555e+00	641	-1.364e-03
642	6.905e+00	642	-1.175e-03
643	3.335e+00	643	-3.833e-04
644	0.000e+00	644	1.157e-03
645	0.000e+00	645	1.477e-03
646	0.000e+00	646	7.460e-05
647	0.000e+00	647	9.803e-04
648	6.050e+00	648	-1.253e-03
649	0.000e+00	649	8.724e-04

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
650	3.030e+00	650	-2.266e-03
651	0.000e+00	651	8.973e-04
652	0.000e+00	652	1.839e-03
653	7.274e+01	653	0.000e+00
654	9.305e+00	654	0.000e+00
655	5.681e+01	655	0.000e+00
656	1.370e+02	656	0.000e+00
657	0.000e+00	657	6.869e-04
658	3.559e+01	658	0.000e+00
659	8.541e+01	659	0.000e+00
660	1.574e+01	660	0.000e+00
661	4.918e+01	661	0.000e+00
662	3.806e+01	662	0.000e+00
663	0.000e+00	663	7.333e-04
664	1.316e+01	664	0.000e+00
665	0.000e+00	665	7.273e-03
666	1.267e+02	666	0.000e+00
667	0.000e+00	667	5.270e-03
668	1.238e+02	668	0.000e+00
669	0.000e+00	669	9.175e-03
670	1.688e+02	670	0.000e+00
671	1.017e+02	671	0.000e+00
672	0.000e+00	672	6.042e-03
673	2.532e+02	673	0.000e+00
674	1.168e+02	674	0.000e+00
675	0.000e+00	675	5.830e-03
676	6.135e+01	676	0.000e+00
677	0.000e+00	677	8.661e-03
678	2.481e+02	678	0.000e+00
679	4.499e+02	679	0.000e+00
680	1.674e+02	680	-6.837e-03
681	7.911e+01	681	0.000e+00
682	1.304e+02	682	0.000e+00
683	4.164e+01	683	-4.036e-03
684	4.589e+00	684	-1.411e-03
685	2.947e+01	685	-2.920e-03
686	0.000e+00	686	0.000e+00
687	3.639e+01	687	-3.096e-03
688	0.000e+00	688	1.097e-03
689	1.118e+01	689	-2.113e-03
690	0.000e+00	690	2.506e-03
691	0.000e+00	691	3.284e-03
692	6.001e+01	692	-1.275e-03
693	4.437e+01	693	-1.922e-03
694	0.000e+00	694	0.000e+00
695	2.888e+01	695	-1.923e-03
696	0.000e+00	696	1.188e-03
697	0.000e+00	697	6.398e-03
698	0.000e+00	698	4.302e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
699	0.000e+00	699	7.065e-03
700	0.000e+00	700	8.198e-03
701	0.000e+00	701	6.528e-03
702	0.000e+00	702	1.175e-02
703	0.000e+00	703	7.337e-03
704	0.000e+00	704	5.338e-03
705	0.000e+00	705	8.984e-03
706	0.000e+00	706	3.644e-03
707	0.000e+00	707	3.948e-03
708	0.000e+00	708	5.057e-03
709	0.000e+00	709	7.612e-03
710	0.000e+00	710	1.072e-02
711	0.000e+00	711	4.036e-03
712	0.000e+00	712	1.411e-03
713	0.000e+00	713	2.920e-03
714	3.186e+00	714	0.000e+00
715	0.000e+00	715	3.096e-03
716	6.605e+00	716	-1.097e-03
717	0.000e+00	717	2.113e-03
718	4.287e+01	718	-2.506e-03
719	3.005e+01	719	-3.284e-03
720	0.000e+00	720	1.275e-03
721	0.000e+00	721	1.922e-03
722	6.719e+01	722	0.000e+00
723	0.000e+00	723	1.923e-03
724	2.402e+02	724	0.000e+00
725	2.678e+02	725	0.000e+00
726	1.287e+02	726	0.000e+00
727	1.497e+01	727	0.000e+00
728	8.698e+01	728	0.000e+00
729	1.441e+02	729	0.000e+00
730	1.861e+02	730	0.000e+00
731	9.175e+02	731	0.000e+00
732	2.987e+02	732	0.000e+00
733	3.611e+01	733	0.000e+00
734	2.560e+02	734	0.000e+00
735	5.919e+02	735	0.000e+00
736	6.551e+01	736	0.000e+00
737	3.837e+02	737	0.000e+00
738	7.032e+01	738	0.000e+00
739	2.276e+02	739	0.000e+00
740	1.426e+02	740	0.000e+00
741	4.455e+01	741	0.000e+00
742	4.880e+02	742	0.000e+00
743	4.126e+02	743	0.000e+00
744	5.899e+02	744	0.000e+00
745	2.968e+02	745	0.000e+00
746	1.014e+03	746	0.000e+00
747	2.419e+02	747	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
748	2.505e+02	748	0.000e+00
749	9.956e+02	749	0.000e+00
750	2.608e+00	750	-5.419e-03
751	2.580e-01	751	-3.772e-04
752	1.912e+00	752	-1.229e-04
753	5.316e+00	753	-2.443e-03
754	1.038e+00	754	-6.109e-03
755	2.994e+00	755	-4.195e-03
756	6.220e-01	756	-9.243e-03
757	2.762e+00	757	-5.439e-03
758	1.334e+00	758	-2.134e-03
759	7.460e-01	759	-7.034e-03
760	2.192e+00	760	-8.957e-05
761	3.000e+00	761	-2.253e-03
762	4.178e+00	762	-3.825e-04
763	2.420e+00	763	-6.367e-03
764	8.758e+00	764	-1.940e-03
765	1.212e+00	765	-1.526e-02
766	2.284e+00	766	-1.391e-03
767	8.234e+00	767	-1.546e-03
768	1.375e+02	768	0.000e+00
769	1.280e+01	769	0.000e+00
770	1.058e+02	770	0.000e+00
771	3.029e+02	771	0.000e+00
772	0.000e+00	772	8.606e-04
773	0.000e+00	773	2.336e-04
774	1.298e+02	774	0.000e+00
775	2.898e+01	775	0.000e+00
776	0.000e+00	776	3.702e-05
777	4.003e+01	777	0.000e+00
778	0.000e+00	778	6.027e-04
779	8.901e+00	779	0.000e+00
780	0.000e+00	780	5.819e-03
781	1.176e+02	781	0.000e+00
782	0.000e+00	782	4.406e-03
783	5.612e+01	783	0.000e+00
784	0.000e+00	784	7.454e-03
785	2.341e+02	785	0.000e+00
786	6.207e+01	786	0.000e+00
787	0.000e+00	787	5.882e-03
788	3.377e+02	788	0.000e+00
789	4.619e+01	789	0.000e+00
790	0.000e+00	790	5.827e-03
791	7.651e+01	791	0.000e+00
792	0.000e+00	792	8.725e-03
793	2.995e+02	793	0.000e+00
794	5.027e+02	794	0.000e+00
795	1.592e+02	795	-5.916e-03
796	7.476e+01	796	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
797	1.512e+02	797	0.000e+00
798	4.839e+01	798	-3.017e-04
799	5.332e+00	799	-4.592e-03
800	3.425e+01	800	-5.926e-04
801	0.000e+00	801	4.804e-03
802	4.229e+01	802	-4.540e-04
803	0.000e+00	803	7.432e-03
804	0.000e+00	804	9.558e-04
805	0.000e+00	805	1.962e-03
806	0.000e+00	806	1.385e-02
807	6.974e+01	807	-2.101e-03
808	1.959e+01	808	0.000e+00
809	5.577e+01	809	-5.407e-04
810	0.000e+00	810	6.956e-04
811	4.863e+01	811	-4.032e-03
812	0.000e+00	812	8.364e-03
813	0.000e+00	813	1.534e-04
814	0.000e+00	814	7.701e-03
815	0.000e+00	815	1.161e-02
816	0.000e+00	816	7.996e-03
817	0.000e+00	817	1.628e-02
818	0.000e+00	818	8.804e-03
819	0.000e+00	819	4.433e-03
820	0.000e+00	820	1.882e-02
821	0.000e+00	821	1.984e-03
822	0.000e+00	822	4.875e-03
823	0.000e+00	823	3.660e-03
824	0.000e+00	824	8.615e-03
825	0.000e+00	825	3.887e-03
826	0.000e+00	826	3.017e-04
827	0.000e+00	827	4.592e-03
828	0.000e+00	828	5.926e-04
829	2.092e+01	829	-4.804e-03
830	0.000e+00	830	4.540e-04
831	8.394e+00	831	-7.432e-03
832	6.789e+01	832	-9.558e-04
833	5.766e+01	833	-1.962e-03
834	4.042e+01	834	-1.385e-02
835	0.000e+00	835	2.101e-03
836	0.000e+00	836	0.000e+00
837	0.000e+00	837	5.407e-04
838	1.228e+01	838	-6.956e-04
839	2.969e+02	839	0.000e+00
840	3.245e+02	840	0.000e+00
841	7.378e+01	841	0.000e+00
842	1.151e+01	842	0.000e+00
843	3.908e+01	843	0.000e+00
844	3.616e+02	844	0.000e+00
845	7.142e+01	845	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
846	1.046e+03	846	0.000e+00
847	2.754e+02	847	0.000e+00
848	2.834e+01	848	0.000e+00
849	2.815e+02	849	0.000e+00
850	4.865e+02	850	0.000e+00
851	1.290e+02	851	0.000e+00
852	4.143e+02	852	0.000e+00
853	7.422e+01	853	0.000e+00
854	4.077e+02	854	0.000e+00
855	1.244e+02	855	0.000e+00
856	3.209e+01	856	0.000e+00
857	5.406e+02	857	0.000e+00
858	5.017e+02	858	0.000e+00
859	5.992e+02	859	0.000e+00
860	3.959e+02	860	0.000e+00
861	1.211e+03	861	0.000e+00
862	4.660e+02	862	0.000e+00
863	2.866e+02	863	0.000e+00
864	1.164e+03	864	0.000e+00
865	3.389e+01	865	0.000e+00
866	0.000e+00	866	6.788e-04
867	7.755e+01	867	0.000e+00
868	0.000e+00	868	2.455e-06
869	8.205e+01	869	0.000e+00
870	1.137e+02	870	0.000e+00
871	1.815e+01	871	0.000e+00
872	2.594e+02	872	0.000e+00
873	1.721e+01	873	0.000e+00
874	0.000e+00	874	1.382e-03
875	0.000e+00	875	1.154e-03
876	2.151e+02	876	0.000e+00
877	0.000e+00	877	6.432e-04
878	1.752e+02	878	0.000e+00
879	0.000e+00	879	1.096e-03
880	2.764e+02	880	0.000e+00
881	0.000e+00	881	1.321e-03
882	0.000e+00	882	2.201e-03
883	0.000e+00	883	4.234e-03
884	0.000e+00	884	3.667e-04
885	0.000e+00	885	2.714e-04
886	0.000e+00	886	1.833e-03
887	0.000e+00	887	5.978e-03
888	0.000e+00	888	5.415e-03
889	0.000e+00	889	3.512e-03
890	0.000e+00	890	7.878e-03
891	0.000e+00	891	4.035e-03
892	0.000e+00	892	1.753e-03
893	0.000e+00	893	2.252e-03
894	0.000e+00	894	7.639e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
895	0.000e+00	895	5.276e-03
896	1.844e+02	896	0.000e+00
897	0.000e+00	897	5.834e-03
898	0.000e+00	898	1.822e-03
899	0.000e+00	899	6.789e-03
900	1.167e+02	900	0.000e+00
901	0.000e+00	901	4.673e-03
902	0.000e+00	902	7.056e-03
903	4.269e+02	903	0.000e+00
904	0.000e+00	904	1.315e-02
905	0.000e+00	905	6.971e-03
906	9.596e+01	906	0.000e+00
907	0.000e+00	907	1.043e-02
908	4.038e+02	908	0.000e+00
909	5.582e+02	909	0.000e+00
910	1.515e+02	910	-2.342e-03
911	8.002e+01	911	0.000e+00
912	1.785e+02	912	0.000e+00
913	5.623e+01	913	-3.112e-03
914	0.000e+00	914	0.000e+00
915	3.980e+01	915	-2.298e-03
916	9.800e+00	916	-1.215e-03
917	4.914e+01	917	-2.744e-03
918	0.000e+00	918	6.132e-04
919	1.510e+01	919	-2.327e-03
920	0.000e+00	920	1.500e-03
921	0.000e+00	921	0.000e+00
922	8.103e+01	922	-8.417e-04
923	5.990e+01	923	-1.058e-03
924	0.000e+00	924	0.000e+00
925	3.899e+01	925	-1.903e-03
926	0.000e+00	926	6.706e-04
927	0.000e+00	927	2.752e-03
928	0.000e+00	928	3.210e-03
929	0.000e+00	929	3.314e-03
930	0.000e+00	930	3.392e-03
931	0.000e+00	931	3.231e-03
932	0.000e+00	932	6.597e-03
933	0.000e+00	933	2.983e-03
934	0.000e+00	934	3.252e-03
935	0.000e+00	935	3.555e-03
936	0.000e+00	936	1.922e-03
937	0.000e+00	937	2.241e-03
938	0.000e+00	938	2.842e-03
939	0.000e+00	939	3.455e-03
940	0.000e+00	940	6.029e-03
941	0.000e+00	941	3.112e-03
942	6.418e-01	942	0.000e+00
943	0.000e+00	943	2.298e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
944	0.000e+00	944	1.215e-03
945	0.000e+00	945	2.744e-03
946	1.087e+01	946	-6.132e-04
947	0.000e+00	947	2.327e-03
948	7.808e+01	948	-1.500e-03
949	4.899e+01	949	0.000e+00
950	0.000e+00	950	8.417e-04
951	0.000e+00	951	1.058e-03
952	6.316e+01	952	0.000e+00
953	0.000e+00	953	1.903e-03
954	3.380e+02	954	0.000e+00
955	3.656e+02	955	0.000e+00
956	4.010e+02	956	0.000e+00
957	3.258e+00	957	0.000e+00
958	5.966e+00	958	0.000e+00
959	1.381e+02	959	0.000e+00
960	8.004e-01	960	0.000e+00
961	1.271e+03	961	0.000e+00
962	3.144e+02	962	0.000e+00
963	3.480e+01	963	0.000e+00
964	2.148e+02	964	0.000e+00
965	5.442e+02	965	0.000e+00
966	1.485e+02	966	0.000e+00
967	5.075e+02	967	0.000e+00
968	7.147e+01	968	0.000e+00
969	5.392e+02	969	0.000e+00
970	1.006e+02	970	0.000e+00
971	1.984e+01	971	0.000e+00
972	7.288e+02	972	0.000e+00
973	6.810e+02	973	0.000e+00
974	4.539e+02	974	0.000e+00
975	4.465e+02	975	0.000e+00
976	1.450e+03	976	0.000e+00
977	4.318e+02	977	0.000e+00
978	3.271e+02	978	0.000e+00
979	1.346e+03	979	0.000e+00
980	1.395e+02	980	0.000e+00
981	1.543e+01	981	0.000e+00
982	7.192e+01	982	0.000e+00
983	1.834e+02	983	0.000e+00
984	1.214e+02	984	0.000e+00
985	3.063e+02	985	0.000e+00
986	3.215e+01	986	0.000e+00
987	4.644e+02	987	0.000e+00
988	3.087e+01	988	0.000e+00
989	0.000e+00	989	4.413e-03
990	0.000e+00	990	9.295e-04
991	5.120e+02	991	0.000e+00
992	0.000e+00	992	2.328e-05

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
993	3.066e+02	993	0.000e+00
994	0.000e+00	994	4.361e-03
995	3.090e+02	995	0.000e+00
996	0.000e+00	996	3.959e-03
997	0.000e+00	997	6.803e-03
998	0.000e+00	998	3.552e-03
999	0.000e+00	999	4.937e-05
1000	0.000e+00	1000	5.731e-04

3 Variáveis Duais (Multiplicadores de Lagrange)

Tabela 2: Variáveis duais (y) do problema PILOT

Coordenada y	Valor y (Dual)
1	-1.105e-01
2	-1.337e-01
3	-2.022e-02
4	-1.556e-01
5	-1.390e-01
6	-2.107e-01
7	-1.255e-01
8	-1.364e-01
9	-1.166e-01
10	-1.567e-01
11	-1.128e-01
12	-1.127e-01
13	-1.386e-01
14	-1.361e-01
15	-1.650e-01
16	-1.512e-01
17	-2.490e-01
18	-1.413e+00
19	-9.859e-02
20	-1.312e-01
21	-1.044e-01
22	-1.310e-01
23	-1.481e-02
24	-1.552e-01
25	-1.375e-01
26	-2.031e-01
27	-1.242e-01
28	-1.357e-01
29	-1.103e-01
30	-1.532e-01
31	-1.118e-01
32	-1.123e-01

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Coordenada y	Valor y (Dual)
33	-1.380e-01
34	-1.224e-01
35	-1.644e-01
36	-1.510e-01
37	-2.484e-01
38	-1.412e+00
39	-6.419e-02
40	-1.283e-01
41	-2.775e-01
42	-4.503e-01
43	-1.631e-01
44	-1.272e-01
45	-2.976e-01
46	-5.043e+00
47	-1.710e+01
48	-6.029e+00
49	-3.333e-03
50	-4.486e-03
51	-3.254e-02
52	-2.196e-02
53	0.000e+00
54	0.000e+00
55	0.000e+00
56	0.000e+00
57	0.000e+00
58	-1.572e-01
59	-1.808e-01
60	-1.593e-01
61	-1.400e-01
62	0.000e+00
63	0.000e+00
64	0.000e+00
65	-1.704e-01
66	0.000e+00
67	-1.032e+00
68	-4.799e-01
69	-6.426e-01
70	-1.313e-02
71	-1.767e+00
72	-9.908e-03
73	-1.379e-02
74	-4.503e-01
75	-2.976e-01
76	-6.029e+00
77	-4.503e-01
78	-7.132e-02
79	0.000e+00
80	0.000e+00
81	0.000e+00

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Coordenada y	Valor y (Dual)
82	-1.593e-01
83	-2.026e+02
84	-1.861e-01
85	-3.741e-01
86	-2.719e-01
87	-3.354e-01
88	-2.496e-01
89	-4.821e-01
90	-3.535e-01
91	-1.103e-01
92	-7.058e-02
93	-3.109e-01
94	-1.968e-01
95	-2.732e+00
96	-2.635e+00
97	-8.402e-01
98	-9.661e-01
99	-2.493e+00
100	-2.493e+00
101	-1.749e+00
102	-1.749e+00
103	-1.749e+00
104	-1.749e+00
105	-4.503e-01
106	-2.976e-01
107	-6.029e+00
108	-2.470e-01
109	-1.062e-01
110	-1.350e-01
111	-2.022e-01
112	-3.805e-01
113	-1.561e-01
114	0.000e+00
115	-4.503e-01
116	0.000e+00
117	-2.063e-03
118	-8.752e-02
119	-2.429e-01
120	-1.634e-01
121	-5.025e-03
122	-2.389e-02
123	-1.671e-01
124	-2.389e-02
125	-1.382e-01
126	-3.362e-01
127	0.000e+00
128	0.000e+00
129	0.000e+00
130	-4.779e-03

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Coordenada y	Valor y (Dual)
131	-1.199e-03
132	0.000e+00
133	-3.616e-02
134	-3.616e-02
135	5.109e-02
136	8.313e-01
137	-1.940e-01
138	-3.180e-03
139	-3.186e-02
140	-7.363e-02
141	-6.171e-02
142	-3.044e-02
143	-7.086e-02
144	-6.329e-02
145	-8.685e-02
146	-1.016e-01
147	-6.892e-02
148	-7.878e-02
149	-1.228e-01
150	-4.420e-02
151	-6.976e-02
152	-7.114e-02
153	-9.173e-02
154	-6.421e-02
155	-8.630e-02
156	-4.141e-02
157	-6.695e-02
158	-6.659e-02
159	-2.037e-02
160	-3.811e-02
161	-2.524e-02
162	-6.401e-03
163	-5.625e-02
164	-4.531e-02
165	-4.661e-02
166	-4.572e-02
167	-4.242e-02
168	-4.710e-02
169	-7.870e-02
170	-3.837e-02
171	-4.578e-02
172	-4.453e-02
173	-5.732e-02
174	-4.998e-02
175	-2.387e-02
176	-2.827e-02
177	-5.282e-02
178	-1.288e-02
179	-9.909e-02

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Coordenada y	Valor y (Dual)
180	-2.063e-01
181	-3.713e-01
182	-1.482e-01
183	-3.485e-01
184	-3.713e-01
185	-3.731e+00
186	-1.103e+01
187	-4.203e+00
188	-3.123e-03
189	-4.486e-03
190	-2.800e-02
191	0.000e+00
192	0.000e+00
193	0.000e+00
194	0.000e+00
195	0.000e+00
196	0.000e+00
197	-5.931e-02
198	-6.284e-02
199	-6.805e-02
200	-5.059e-02
201	0.000e+00
202	0.000e+00
203	4.911e-02
204	-4.911e-02
205	-4.911e-02
206	-4.260e-02
207	-1.633e-01
208	-5.180e-01
209	-5.988e-01
210	-4.480e-01
211	-1.217e-02
212	-1.232e+00
213	-9.339e-03
214	-1.278e-02
215	-3.713e-01
216	-3.713e-01
217	-4.203e+00
218	-3.713e-01
219	-6.836e-02
220	0.000e+00
221	0.000e+00
222	0.000e+00
223	-6.805e-02
224	-4.797e-02
225	-5.630e-01
226	-4.041e-01
227	-4.555e-01
228	-3.840e-01

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Coordenada y	Valor y (Dual)
229	-5.779e-01
230	-1.178e+00
231	-6.200e-01
232	-1.158e+00
233	-7.080e-01
234	-8.289e-01
235	-1.589e+00
236	-4.203e+00
237	-4.203e+00
238	-3.713e-01
239	-9.909e-02
240	0.000e+00
241	-1.256e-01
242	0.000e+00
243	-1.290e+00
244	-1.285e+00
245	-8.427e-01
246	-8.456e-01
247	-9.185e-01
248	-9.368e-01
249	-2.101e+00
250	-2.116e+00
251	-5.885e-01
252	-5.819e-01
253	-7.931e-01
254	-7.771e-01
255	-6.547e-01
256	-6.856e-01
257	-4.913e-01
258	-6.362e-01
259	-5.291e-01
260	-5.291e-01
261	-1.219e+00
262	-1.219e+00
263	-1.219e+00
264	-1.219e+00
265	-3.713e-01
266	-3.713e-01
267	-4.203e+00
268	-6.448e-02
269	-2.045e-02
270	-2.279e-02
271	-4.398e-02
272	-2.403e-02
273	-2.020e-02
274	0.000e+00
275	-3.713e-01
276	0.000e+00
277	-3.513e-03

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Coordenada y	Valor y (Dual)
278	-8.010e-02
279	-2.059e-01
280	-1.634e-01
281	-4.003e-03
282	-2.199e-02
283	-1.512e-01
284	-2.389e-02
285	-1.434e-01
286	-1.649e-02
287	0.000e+00
288	-4.743e-02
289	-2.399e-01
290	-3.807e-03
291	-8.334e-04
292	-3.155e-03
293	-1.257e-02
294	-1.257e-02
295	2.622e-02
296	-2.125e+00
297	-4.479e-01
298	-2.971e-03
299	-1.361e-02
300	-4.332e-02
301	-3.725e-02
302	-3.343e-02
303	-4.406e-02
304	-5.859e-02
305	-4.570e-02
306	-5.302e-02
307	-4.628e-02
308	-4.165e-02
309	-5.859e-02
310	-4.687e-02
311	-4.916e-02
312	-5.462e-02
313	-4.343e-02
314	-4.568e-02
315	-4.009e-02
316	-4.174e-02
317	-4.845e-02
318	-4.797e-02
319	-1.963e-02
320	-2.367e-02
321	-1.866e-02
322	-5.291e-03
323	-3.552e-02
324	-3.392e-02
325	-2.727e-02
326	-2.599e-02

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Coordenada y	Valor y (Dual)
327	-2.754e-02
328	-2.678e-02
329	-4.074e-02
330	-2.645e-02
331	-3.035e-02
332	-2.722e-02
333	-2.009e-02
334	-2.049e-02
335	-1.459e-02
336	-2.117e-02
337	-2.107e-02
338	-1.156e-02
339	-4.871e-02
340	-2.147e-01
341	-2.627e-01
342	-1.640e-01
343	-1.512e-01
344	-1.682e-01
345	-1.684e+00
346	-3.680e+00
347	-1.928e+00
348	-1.623e-03
349	-2.148e-03
350	-1.433e-02
351	0.000e+00
352	0.000e+00
353	0.000e+00
354	0.000e+00
355	0.000e+00
356	0.000e+00
357	-4.350e-02
358	-4.866e-02
359	-5.465e-02
360	-5.334e-02
361	0.000e+00
362	0.000e+00
363	5.859e-02
364	-5.859e-02
365	-5.859e-02
366	-4.045e-02
367	-4.898e-01
368	-1.826e-01
369	-2.714e-01
370	-2.054e-01
371	-8.579e-03
372	-5.649e-01
373	-6.486e-03
374	-9.030e-03
375	-2.627e-01

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Coordenada y	Valor y (Dual)
376	-1.682e-01
377	-1.928e+00
378	-2.627e-01
379	-4.760e-02
380	-9.078e-02
381	0.000e+00
382	0.000e+00
383	-5.465e-02
384	-4.314e-02
385	-1.642e-01
386	-2.059e-01
387	-1.826e-01
388	-1.377e-01
389	-2.670e-01
390	-5.222e-01
391	-2.142e-01
392	-4.787e-01
393	-2.319e-01
394	-3.749e-01
395	-1.246e-01
396	-1.928e+00
397	-1.928e+00
398	-1.682e-01
399	-4.871e-02
400	0.000e+00
401	-2.012e-01
402	0.000e+00
403	-1.661e-01
404	-1.652e-01
405	-1.258e-01
406	-1.286e-01
407	-1.292e-01
408	-1.314e-01
409	-2.833e-01
410	-2.898e-01
411	-7.665e-02
412	-7.635e-02
413	-1.091e-01
414	-1.069e-01
415	-2.690e-01
416	-4.206e-01
417	-4.261e-01
418	-2.517e-01
419	-2.398e-01
420	-2.398e-01
421	-5.590e-01
422	-5.590e-01
423	-5.590e-01
424	-5.590e-01

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Coordenada y	Valor y (Dual)
425	-2.627e-01
426	-1.682e-01
427	-1.928e+00
428	-4.994e-02
429	-1.727e-02
430	-2.472e-02
431	-2.929e-02
432	-2.681e-02
433	-2.664e-02
434	-2.929e-02
435	-4.448e-02
436	0.000e+00
437	-1.501e-03
438	-7.181e-02
439	-2.038e-01
440	-1.634e-01
441	-9.199e-04
442	-1.915e-02
443	-1.045e-01
444	-1.915e-02
445	-1.067e-01
446	-5.516e-01
447	-4.737e-01
448	0.000e+00
449	-5.253e-03
450	-2.720e-03
451	-2.848e-04
452	0.000e+00
453	-9.731e-03
454	-9.731e-03
455	1.631e-02
456	-6.867e-01
457	-1.860e-01
458	-1.441e-03
459	-1.093e-02
460	-3.420e-02
461	-2.824e-02
462	-2.903e-02
463	-3.401e-02
464	-4.573e-02
465	-3.573e-02
466	-4.343e-02
467	-3.615e-02
468	-3.319e-02
469	-4.645e-02
470	-2.240e-02
471	-2.413e-02
472	-2.628e-02
473	-3.368e-02

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Coordenada y	Valor y (Dual)
474	-3.544e-02
475	-3.396e-02
476	-3.294e-02
477	-3.701e-02
478	-3.389e-02
479	-1.548e-02
480	-1.840e-02
481	-1.337e-02
482	-3.933e-03
483	-2.708e-02
484	-2.625e-02
485	-2.070e-02
486	-2.012e-02
487	-2.106e-02
488	-2.003e-02
489	-3.141e-02
490	-1.649e-02
491	-1.780e-02
492	-1.860e-02
493	-1.492e-02
494	-1.549e-02
495	-1.095e-02
496	-1.646e-02
497	-1.539e-02
498	-8.837e-03
499	-3.812e-02
500	-1.500e-01